

Principle of antimicrobial therapy

PYPH305 Pharmacology 2 (ภาคปว ๓๐๕ เกสัชวิทยา 2)

Inhibition of Protein, Nucleic, DNA synthesis

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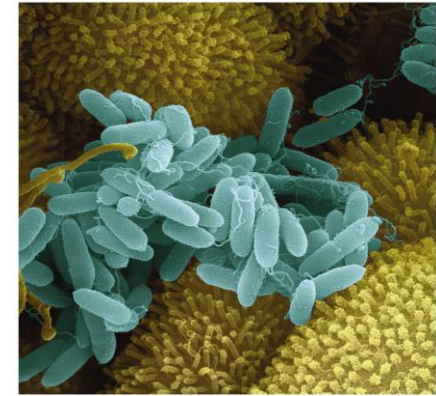
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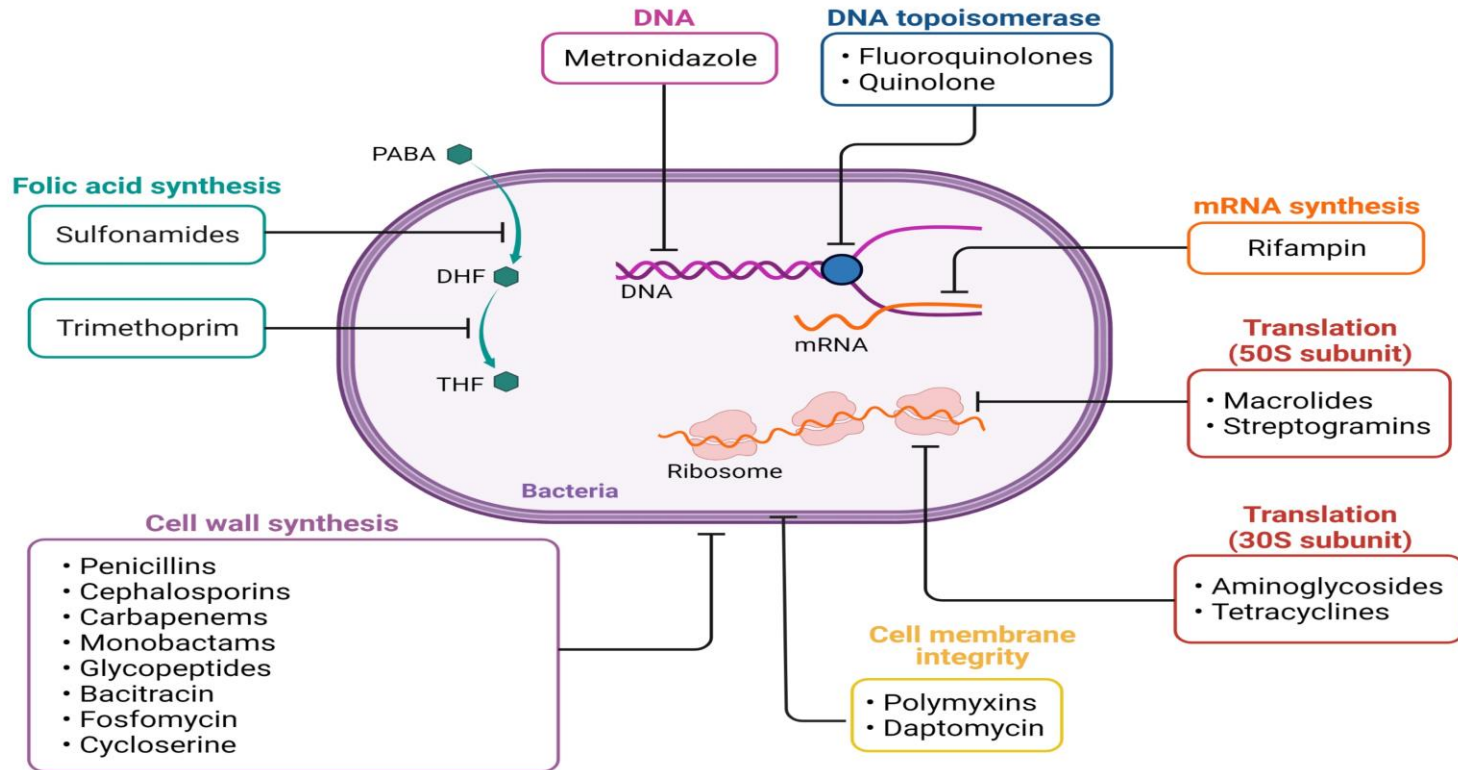
Antibiotic Drugs

3. DNA/RNA synthesis inh.



Fluoroquinolones

- Novel group of synthetic antibiotics developed in response to growing resistance
- Agents available today are all structural derivatives of **nalidixic acid**
- The *fluorinated* quinolones (FQs) represent a major therapeutic advance:
 - **Broad spectrum of activity**
 - **Improved PK properties – excellent bioavailability, tissue penetration, prolonged half-lives**
 - **Overall safety**
- Disadvantages: emergence of resistance



Fluroquinolones

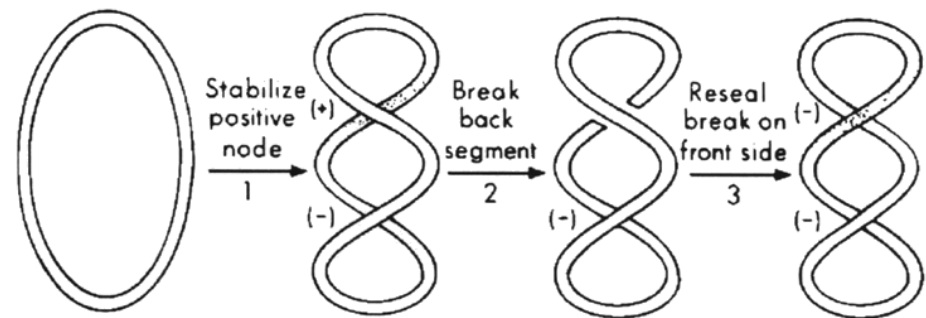
- -floxacin
- Examples
 - 1st nalidixic acid
 - 2nd Nor, Olf, Cipro, Ruflo
 - 3rd Levo
 - 4th Moxi, Sitaflo, Pru
- MOA
 - Bind to DNA gyrase G(-)
 - Bind to Topoisomerase IV G (+)
- Result: Blocks DNA
 - Repair
 - Replication
 - Transcription
- Bactericidal
- Broad Spectrum
 - Gr – ve
 - Gr + ve 3th and 4th gen
 - Atypical bacteria
 - Mycobacterium (MTB)
 - Mycobacterium avium complex (MAC)

Older FQs

- Norfloxacin (Noroxin®) PO
- Ciprofloxacin (Cipro®) PO, IV
- Ofloxacin (Floxin®) PO, IV

Newer FQs

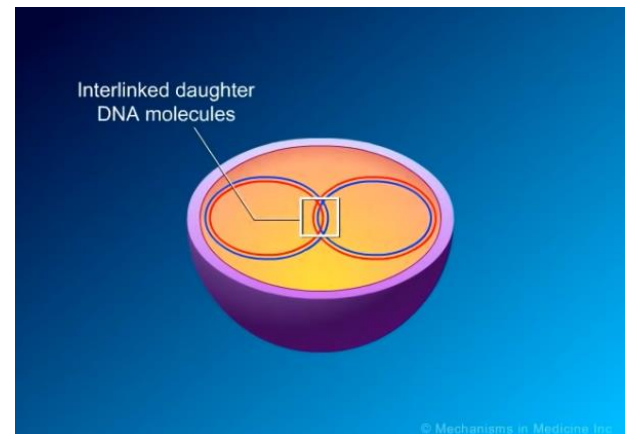
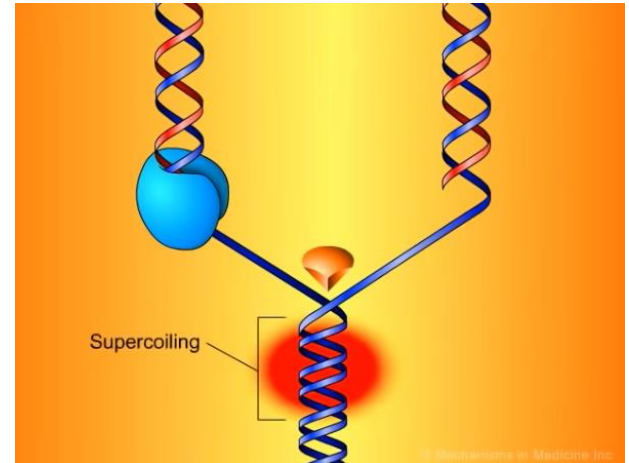
- Levofloxacin (Levaquin®) PO, IV
- Moxifloxacin (Avelox®) PO, IV
- Gatifloxacin (Tequin®) Eye Drop
- Gemifloxacin (Factive®) PO
- Sitafloxacina (Gracevit®) PO

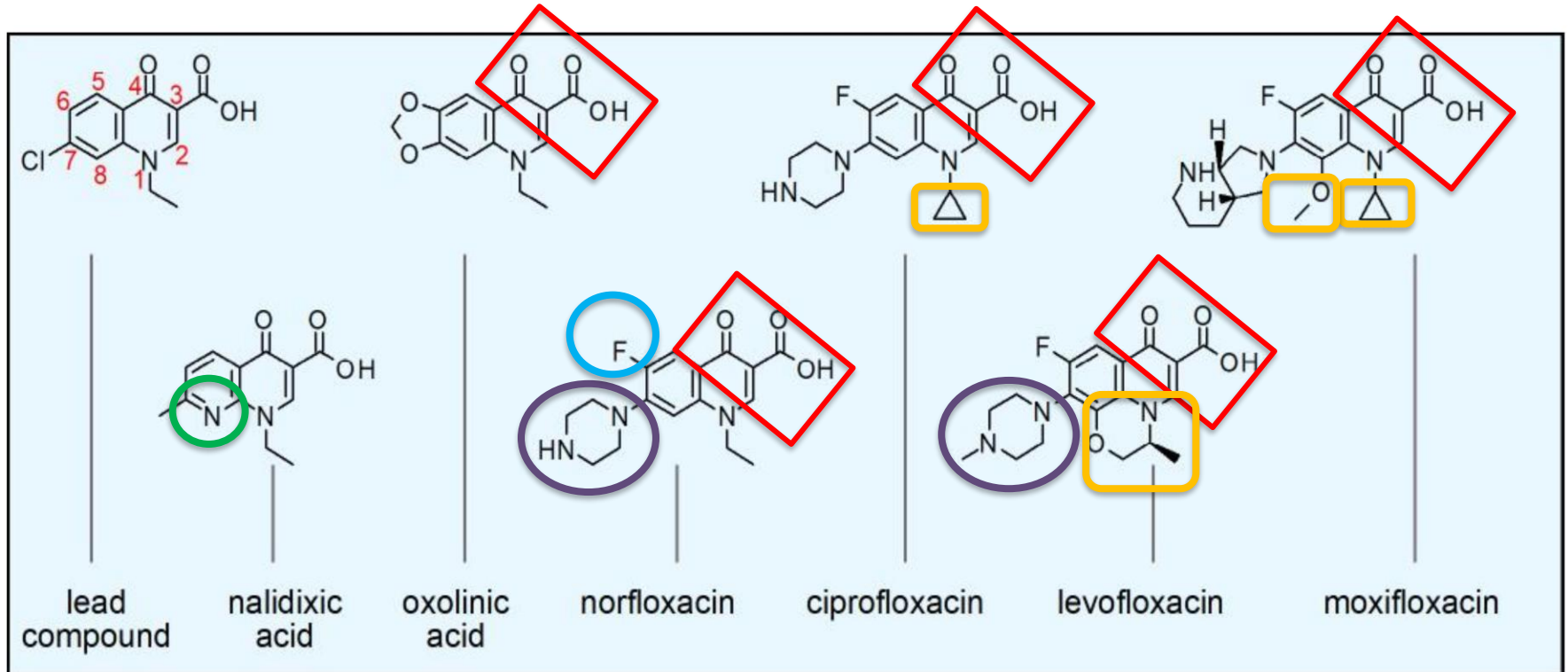


Fluoroquinolones

Mechanism of Action

- Unique mechanism of action
- Inhibit bacterial topoisomerases which are necessary for DNA synthesis
 - *DNA gyrase* – removes excess positive supercoiling in the DNA helix
 - Primary target in gram-negative bacteria
 - *Topoisomerase IV* – essential for separation of interlinked daughter DNA molecules
 - Primary target for many gram-positive bacteria
- FQs display **concentration-dependent bactericidal activity**

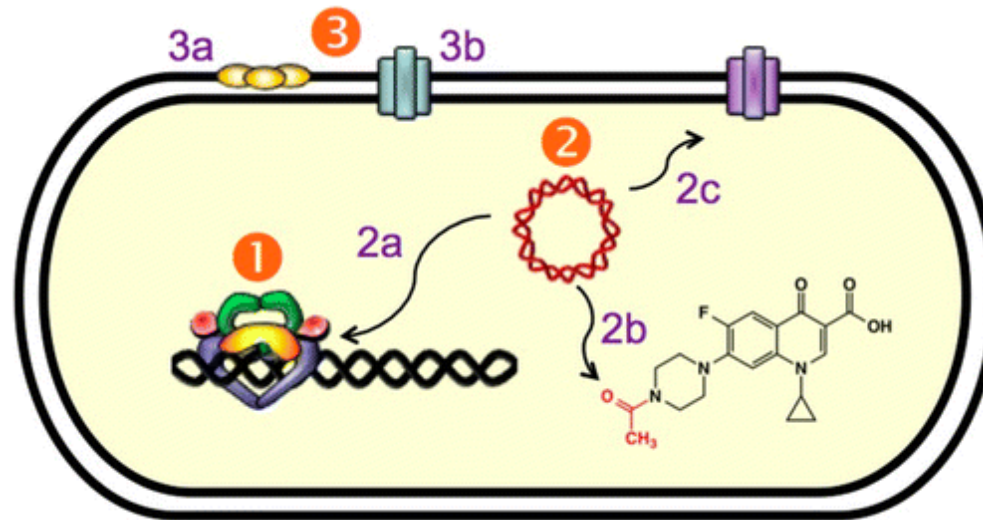




generation 0	1st generation	2nd generation	3rd generation	4th generation
no clinical use	mostly not in use	most of the introduced molecules remain in use		
Gram spectrum				
-	-	- and some +	- and more +	- and many+
potency				
→				

- Bind to DNA and topoisomerase II, IV
- Decrease MIC
- High protein binding
- Longer half life, Gram+
- Broad Gram -

Fluoroquinolones: Mechanisms of Resistance



1. Target-mediated quinolone resistance

- Mutations in gyrase and topoisomerase IV weaken quinolone– enzyme interactions
- Disruption of the water–metal ion bridge (quinolone resistance determining region (QRDR))

2. Plasmid-mediated quinolone resistance

- (2a) Qnr proteins (yellow) decrease topoisomerase–DNA binding and protect enzyme– DNA complexes from quinolones
- (2b) Aac(6')-Ib-cr is an aminoglycoside acetyltransferase that acetylates the free nitrogen on the C7 ring of ciprofloxacin and norfloxacin, decreasing their effectiveness.
- (2c) Plasmid-encoded efflux pumps decrease the concentration of quinolones in the cell.

3. Chromosome-mediated quinolone resistance

- (3a) Underexpression of porins in Gram-negative species decreases drug uptake.
- (3b) Overexpression of chromosome-encoded efflux pumps decreases drug retention in the cell.

FQs Spectrum of Activity

Gram-positive – older agents with poor activity; newer FQs with enhanced potency

- Methicillin-susceptible *Staphylococcus aureus*
- *Streptococcus pneumoniae* (including PRSP) – Levo, Moxi, Gemi
- Group and viridans streptococci and *Enterococcus sp.* – limited activity

Gram-Negative – most FQs have excellent activity

(cipro=levo>gati>moxi)

- *Enterobacteriaceae* – including *E. coli*, *Klebsiella sp*, *Enterobacter sp*, *Proteus sp*, *Salmonella*, *Shigella*, *Serratia marcescens*, etc.
- *H. influenzae*, *M. catarrhalis*, *Neisseria sp*.
- *P aeruginosa* – significant resistance has emerged; ciprofloxacin and levofloxacin with best activity

FQs Spectrum of Activity

Anaerobes – moxifloxacin

Atypical Bacteria – all FQs have excellent activity against atypical bacteria including:

- *Legionella pneumophila* – Drug of choice
- *Chlamydia sp.*
- *Mycoplasma sp.*
- *Ureaplasma urealyticum*

Other Bacteria – *Mycobacterium tuberculosis*, *Bacillus anthracis*

Fluoroquinolones: Pharmacology

- Concentration-dependent bacterial killing – AUC/MIC correlates with efficacy
- Absorption
 - Most FQs have good bioavailability after oral administration (not norfloxacin)
 - C_{max} within 1 to 2 hours; co-administration with food delays the peak concentration
- Distribution
 - Extensive tissue distribution – prostate; liver; lung; skin/soft tissue and bone; urinary tract (cipro, levo, gati)
 - Minimal CSF penetration
- Elimination – renal and hepatic; not removed by Hemodialysis
- Half-life
 - Short – Nor, Oflo, Cipro, Ruflo, Pruli, Sitaflo
 - Long – Levo, Moxi

Fluoroquinolones: Clinical Uses

- Upper Respiratory Tract Infections
 - Sinusitis, Acute bacterial exacerbations of chronic bronchitis (ABECB) – *levo, moxi, gemi*
- Lower Respiratory Tract Infections
 - Community-acquired pneumonia - *levo, moxi*
 - Nosocomial pneumonia – *cipro, levo*
 - Bacterial exacerbations in cystic fibrosis - *cipro*
- Urinary Tract Infections
 - Cystitis, pyelonephritis, prostatitis – *cipro, levo,*
- Skin infections
- Other: osteo, intraabdominal, Gastroenteritis, STDs, TB

Fluoroquinolones: Clinical Uses

- Norfloxacin
 - Tablet, not reach the systemic level
 - Indications – gastroenteritis, gonorrhea
- Rufloxacin
 - Tablet
 - Indications – UTIs and prostatitis
- Ciprofloxacin
 - Tablet, IV infusion, eye drop
 - Most active against g(-) (*P.aeruginosa*)
 - High incidence of arthropathy
 - Strong CYP1A2 inhibitor
 - Indications – Usually keep for *P.aeruginosa*

Fluoroquinolones: Clinical Uses

- Ofloxacin (racemic mixture)
 - Tablet, eye drop, ear drop
 - Indications – UTIs, URIs (otitis media), LRIs, SSSIs gastroenteritis and MTB
- Levofloxacin (S-enantiomer of ofloxacin)
 - Tablet, IV infusion, eye drop
 - Better anti-bacterial activity
 - Active against g(+) (*S.pneumoniae*)
 - Indications – UTIs, URIs, LRIs, SSSIs, *H.pylori* eradication and others

Fluoroquinolones: Clinical Uses

- Gatifloxacin
 - eye drop
 - Systemic use was withdrawn (severe diabetes)
 - Indications – bacterial conjunctivitis
- Moxifloxacin
 - Tablet, IV infusion
 - Active against G (+) (*S.pneumoniae*)
 - Non renal excretion
 - Indications – URIs, LRIs and skin and skin structure infections (SSSIs) (not for UTIs) and MTB
- Prulifloxacin, Sitafloxacin
 - Tablet take twice a day
 - Indications – URIs , LRIs, UTIs, SSSIs and others

Fluoroquinolones: Adverse Effects

- Gastrointestinal – 5 %
 - Nausea, vomiting, diarrhea, dyspepsia
- Central Nervous System
 - Headache, agitation, insomnia, dizziness, rarely, hallucinations and **seizures in elderly**
- Peripheral neuropathy
- Hepatotoxicity
 - LFT elevation (led to withdrawal of trovafloxacin)
- Phototoxicity (uncommon with current FQs)
 - More common with older FQs (halogen at position 8)

Fluoroquinolones: Adverse Effects

- **Articular Damage**
 - Arthropathy including articular cartilage damage, arthralgias, and joint swelling
 - Contraindication in pediatric patients and pregnant or breastfeeding women
 - Risk versus benefit
- **Cardiac**
 - Risk of aortic aneurysm or dissection such as those with peripheral atherosclerotic vascular disease, hypertension, certain genetic conditions such as Marfan syndrome and Ehlers-Danlos syndrome, and the elderly.
 - Variable prolongation in QTc interval (Led to withdrawal of grepafloxacin, sparfloxacin)
- **Other adverse reactions: tendon rupture, dysglycemias (gatifloxacin), hypersensitivity**

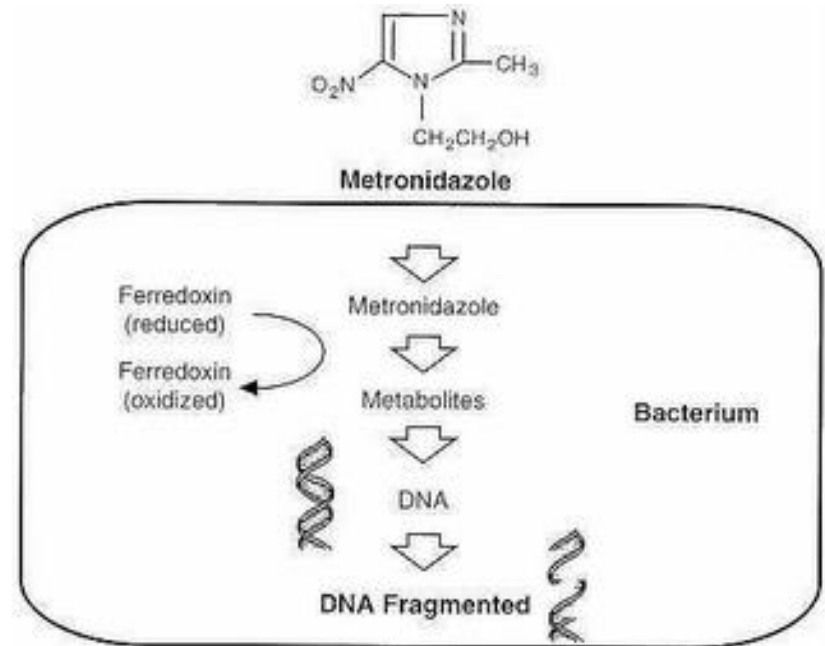
Fluoroquinolones: Drug Interactions

- **Divalent and trivalent cations – ALL FQs**
 - Zinc, Iron, Calcium, Aluminum, Magnesium
 - Antacids, Sucralfate, ddi, enteral feedings
 - Impair oral absorption of orally-administered FQs
 - may lead to CLINICAL FAILURE
 - Administer doses 2 to 4 hours apart; FQ first
- Theophylline and Cyclosporine - cipro
 - Inhibition of metabolism, ↑ levels, ↑ toxicity
- Warfarin – idiosyncratic, all FQs

Mechanism of Action

- Ultimately inhibits DNA synthesis
 - Prodrug which is activated by a reductive process
 - Selective toxicity against **anaerobic** and microaerophilic bacteria due to **the presence of ferredoxins within these bacteria**
 - **Ferredoxins donate electrons** to form highly reactive nitro anion which damage bacterial DNA and cause cell death
- Metronidazole displays **concentration-dependent bactericidal activity**

Metronidazole



Mechanisms of Resistance –
Impaired **oxygen scavenging** ability –
higher local oxygen concentrations
which decreases activation of
metronidazole

- Altered ferredoxin levels –
reduced transcription of the
ferredoxin gene; less activation of
metronidazole

Metronidazole: Spectrum of Activity

Anaerobic Bacteria (Below Diaphragm)

Bacteroides sp. (ALL)

Fusobacterium

Prevotella sp.

Clostridium sp. (ALL)

Helicobacter pylori

Anaerobic Protozoa

Trichomonas vaginalis

Entamoeba histolytica

Giardia lamblia

Gardnerella vaginalis

Metronidazole: Pharmacology

Absorption – available IV and PO

- Rapidly and completely absorbed ($F > 90\%$); food with minimal effect on absorption

Distribution

- Good serum concentrations with PO or IV
- Well absorbed into body tissues and fluids; DOES penetrate the CSF

Elimination

- Metronidazole is primarily metabolized by the liver (metabolites excreted in urine); half-life is 6 to 8 hrs
- Metronidazole is removed during hemodialysis

Metronidazole: Clinical Uses

- Anaerobic Infections (**including in the CNS**)
 - Intraabdominal, pelvic, skin/soft tissue, diabetic foot and decubitus ulcer infections; brain abscess
- **Pseudomembranous colitis due to *C. difficile***
 - **Metronidazole is the DRUG OF CHOICE**
 - **PO or IV**
- Other
 - Bacterial vaginosis, Trichomonas, Amebiasis, *H. pylori*, Rosacea, Gingivitis, Giardia

Metronidazole: Adverse Effects

- Gastrointestinal
 - Nausea, vomiting, stomatitis, metallic taste
- CNS – most serious
 - Peripheral neuropathy, seizures, encephalopathy
 - Use with caution in patients with preexisting CNS disorders
 - Requires discontinuation of metronidazole
- Disulfuram-like reactions (aldehyde dehydrogenase ALDH inhibition)
- Potentiation of warfarin effect (CYP2C9 inhibition)
- Mutagenicity, carcinogenicity
 - Avoid during pregnancy and breastfeeding

Antibiotic Drugs 4. Protein synthesis inh.

- Bacteriostatic
- Their spectrum of activity is limited to **Gram-positive** cocci such as *Streptococci* and *Staphylococci*.
- These antibiotics are also active against anaerobes.
- Four different classes of antibiotics which are unrelated in terms of structure but which have a similar mode of action and spectrum of activity.

MLSK: Protein synthesis inhibitors

Ketolides
Telithromycin

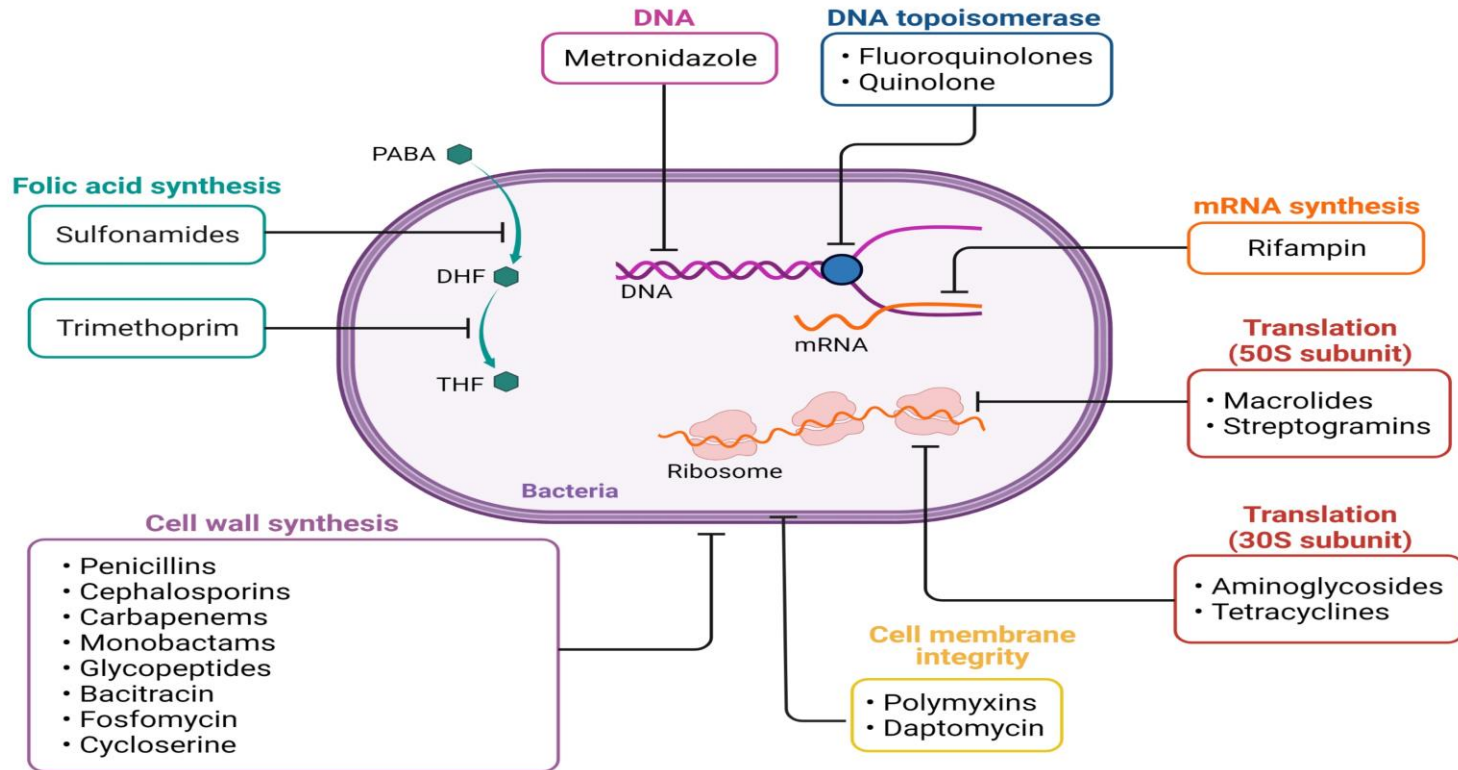
50S Inhibition

Mode of Action
of
MLSK

Clindamycin
Lincosamides

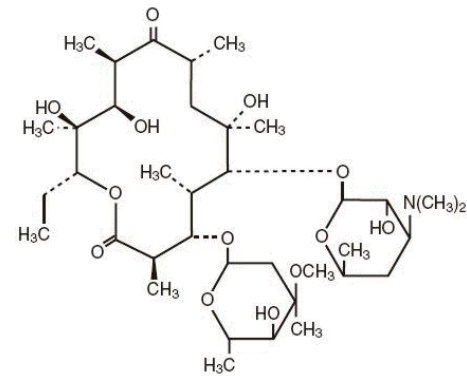
Quinupristin-Dalfopristin
Streptogramins

Erythromycin
Macrolides

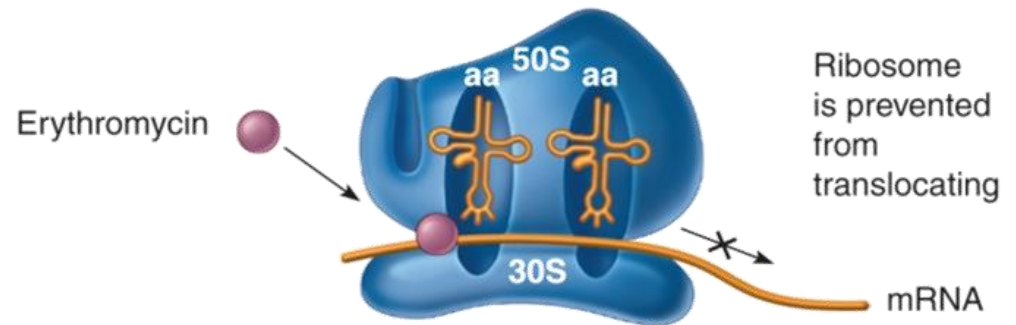


Macrolides

- -mycin (usually)
- *Actinomyces* source
- Examples
 - Erythromycin
 - Azithromycin
 - Clarithromycin
 - Roxithromycin
 - Midecamycin
 - Spiramycin
- MOA
 - 50S reversible binding
 - Inhibits elongation
 - Blocks translocation
- Result
 - Inhibit protein synthesis
- G(+) and select G(-)

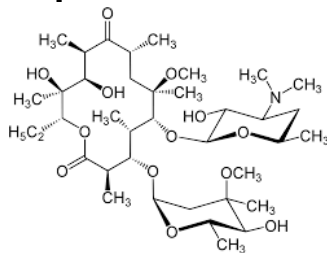


Erythromycin

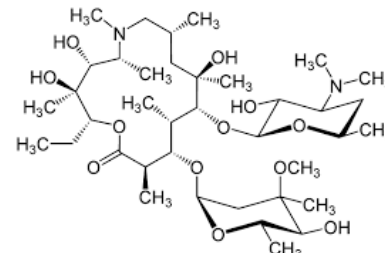


Macrolides

- Erythromycin is a naturally-occurring macrolide derived from *Streptomyces erythreus* – problems with **acid lability**, narrow spectrum, poor GI intolerance, short elimination half-life
- Structural derivatives include clarithromycin and azithromycin:
 - Broader spectrum of activity
 - Improved PK properties – better bioavailability, better tissue penetration, prolonged half-lives
 - Improved tolerability



Clarithromycin



Azithromycin

Macrolides and Ketolides

Mechanism of Action

- Inhibits protein synthesis by reversibly binding to the 50S ribosomal subunit
 - Suppression of RNA-dependent protein synthesis
 - **Telithromycin (Ketek®) binds to 2 domains** on 50S ribosome (10 times stronger binding to domain II – may account for greater spectrum of activity)
- Macrolides and ketolides typically display bacteriostatic activity, but may be bactericidal when present at high concentrations against very susceptible organisms
- **Time-dependent activity with PAE**

Macrolides and Ketolides

Mechanisms of Resistance

- **Active efflux** (accounts for 80% in US) – *mef* gene encodes for an **efflux pump** which pumps the macrolide out of the cell away from the ribosome; confers *low level* resistance to macrolides and ketolides
- **Altered target sites** (primary resistance mechanism in Europe) – **encoded by the *erm* gene** which alters the macrolide binding site on the ribosome; confers *high level* resistance to all macrolides, clindamycin and Synercid, **but telithromycin retains activity**
- **Modification of ribosome binding site**
 - Methylation (constitutive or institutive)
 - MLSB (macrolide – lincosamide – streptogramin B)
 - Cross-resistance occurs between all macrolides

Macrolide and Ketolide: Spectrum of Activity

Gram-Positive Aerobes – telithromycin, erythromycin and clarithromycin display the best activity

(Telithro>Clarithro>Erythro>Azithro)

- Methicillin-susceptible *Staphylococcus aureus*
- *Streptococcus pneumoniae* (resistance is emerging) –
 - telithromycin has activity against macrolide-resistant *Streptococcus pneumoniae*
- Group and viridans streptococci
- *Bacillus sp.*, *Corynebacterium sp.*

Gram-Negative Aerobes – newer macrolides with enhanced activity, ketolides with poor activity

(Azithro>Clarithro>Erythro> Telithro)

- *H. influenzae* (not erythro or telithro), *M. catarrhalis*, *Neisseria sp.*
- Do NOT have activity against any *Enterobacteriaceae*

Macrolide and Ketolide: Spectrum of Activity

Anaerobes – activity against upper airway anaerobes

Atypical Bacteria – all macrolides have excellent activity against atypical bacteria including:

- *Legionella pneumophila* - DOC
- *Chlamydia sp.*
- *Mycoplasma sp.*
- *Ureaplasma urealyticum*

Other Bacteria – *Mycobacterium avium complex* (MAC – only A and C), *Treponema pallidum*, *Campylobacter*, *Borrelia*, *Bordetella*, *Brucella*. *Pasteurella*

Macrolides: Pharmacology

Absorption

- Erythromycin – variable absorption ($F = 15-45\%$); food may decrease the absorption
 - Base: destroyed by gastric acid; enteric coated
 - Esters and ester salts: more acid stable
- Clarithromycin – acid stable and well-absorbed ($F = 55\%$) regardless of presence of food
- Azithromycin – acid stable; $F = 38\%$; food decreases absorption of capsules
- Telithromycin – only available PO; $F = 57\%$

Macrolides and Ketolides: Pharmacology

Distribution

- **Extensive tissue and cellular distribution** – clarithromycin and azithromycin with extensive tissue penetration
- **Minimal CSF penetration**

Elimination

- Clarithromycin is the only macrolide partially eliminated by the kidney (18% of parent and all metabolites); requires dose adjustment when $\text{CrCl} < 30 \text{ ml/min}$
- Hepatically eliminated: ALL
- **NONE of the macrolides are removed during hemodialysis!**
- Variable elimination half-lives (1.4 hours for erythro; 3 to 7 hours for clarithro; **68 hours for azithro**, 10 hours for telithro)

Macrolides and Ketolides: Clinical Uses

- Respiratory Tract Infections
 - Pharyngitis/ Tonsillitis – pen-allergic patients (telithro not approved)
 - Sinusitis, acute bacterial exacerbations of chronic bronchitis (ABECB) (azithro best if *H. influenzae* suspected), Otitis Media
 - Community-acquired pneumonia - atypical
- Uncomplicated Skin & Soft Tissue Infections – Clarithro, Erythro, Azithro
- STDs, Gonococcal urethritis and cervicitis – Single 2 gram dose of azithro
- MAC – Azithro for proph; Clarithro for RX

Macrolides and Ketolides: Clinical Uses

Erythromycin

- Dosage form – Ery base (acid labile), stearate (tab), estolate (Cap, Syr), ethylsuccinate (Syr)
- Other actions – **Motilin receptor agonist**
- Indications – Corynebacterial infections, URIs, Intestinal paralysis
- PK: serum half life = 1.5 hours, Potent CYP3A4 inhibitor, excreted in bile and lost in feces (95%)
- Adverse events – Abdominal cramp, anorexia, N/V, diarrhea, Liver toxic (Ery estolate), May prolong QT-interval

Macrolides and Ketolides: Clinical Uses

Roxithromycin (PO)

- Dosage form – Tablet
- PK: food decrease absorption 21%, serum half life = 10.5 hours, excreted unchanged in bile (65%), **low CYP interaction**
- More active against *Legionella pneumophila*
- Low GI adverse events
- Indications – URIs and skin infection

Macrolides and Ketolides: Clinical Uses

Clarithromycin

- Dosage form – Tablet, MR (modified release), syrup, IV
- PK: acid resistance, active metabolite (14-hydroxy-clarithromycin), **Potent CYP3A4 inhibitor**, excreted by urine
- Adverse events – Low GI side effect, Alteration of sense of smell with taste perversion, may prolong QT-interval
- Indications – *H.pylori* triple therapy (PPIs + amoxicillin+clarithromycin), URIs, LRIs, SSSIs, *Mycobacterium avium complex* (MAC)

Macrolides and Ketolides: Clinical Uses

Azithromycin

- Dosage form – Capsule, Tablet, Powder in sachet, powder for reconstitution, Prolong release granule, IV infusion
- PK: Unique PK, food alter bioavailability of capsule, Concentration – tissue:serum =50:1, $t_{1/2} = 3 \text{ days}$, No CYP inhibition, biliary excretion (without metabolism)
- Indications – more active against *H.influenza* and *Chlamydia spp.*, URIs, LRIs (CAP, AECB), STD, *Mycobacterium avium complex* (MAC)
- **~25% of Strep pneumo is resistant to Azithromycin**, so combine with Ceftriaxone for patients sick enough to hospitalize with community-acquired Pneumonia (or recent abx use).

Macrolides : Adverse Effects

- Gastrointestinal – up to 33 %
 - Nausea, vomiting, diarrhea, dyspepsia
 - Most common with erythro; less with new agents
- Cholestatic hepatitis - rare
 - > 1 to 2 weeks of erythromycin estolate
- Thrombophlebitis – IV Erythro and Azithro
 - Dilution of dose; slow administration
- Other: ototoxicity (high dose erythro in patients with renal impairment); QTc prolongation; allergy

Ketolides: Adverse Effects

- Gastrointestinal
 - Nausea (7%), vomiting (3%), diarrhea (10%)
- Central Nervous System
 - Dizziness (3%), headache (2%)
- Hepatotoxicity – severe liver injury; FDA recently removed 2 indications for use (risk vs benefit)
- Ocular – blurred vision, decreased accommodation
- QTc prolongation

Macrolides and Ketolides

Drug Interactions

Erythromycin, Clarithromycin and Telithromycin – are ***inhibitors*** of cytochrome p450 system in the liver; **may increase** concentrations of:

Theophylline

Digoxin, Disopyramide

Carbamazepine

Valproic acid

Cyclosporine

Terfenadine, Astemizole

Phenytoin

Cisapride

Warfarin

Ergot alkaloids

Tacrolimus

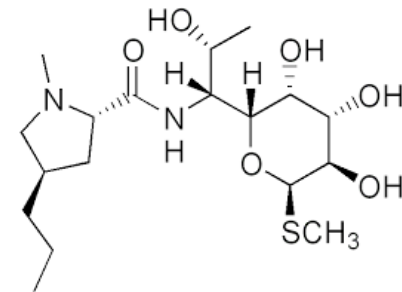
NOT AZITHROMYCIN

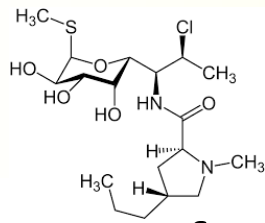
Lincosamides: Lincomycin & Clindamycin

- G(+), some anaerobes
- MOA
 - Bind to P-site of 50S, Block elongation, Inhibits protein synthesis

Lincomycin

- Cap, IV, IM
- Spectrum – Gram (+), anaerobes
- Indications
 - URIs – lincomycin + dexamethasone for tonsillitis (off-labeled indications)
- Adverse events
 - Pseudomembranous colitis
 - Neuromuscular blocking effect





Clindamycin

Mechanism of Action

- Inhibits protein synthesis by binding exclusively to the **50S ribosomal subunit**
 - Binds in close proximity to macrolides – competitive inhibition
- Typically displays bacteriostatic activity, but may be bactericidal when present at high concentrations against very susceptible organisms

Mechanisms of Resistance

- **Altered target sites** – encoded by the *erm* gene which alters the clindamycin binding site on the ribosome; confers *high level* resistance to all macrolides, clindamycin and Synercid (MLS-type B)
- **Active efflux** – *mef* gene encodes for an efflux pump which pumps the macrolide out of the cell but NOT clindamycin; confers *low level* resistance to macrolides, but clindamycin still active

Gram negative are intrinsically resistant

Clindamycin: Spectrum of Activity

Gram-Positive Aerobes

- Methicillin-susceptible *Staphylococcus aureus* (MSSA only)
- *Streptococcus pneumoniae* (only PSSP) – resistance is developing
- Group and viridans streptococci

Anaerobes – activity against Above the Diaphragm Anaerobes (ADA)

Peptostreptococcus some *Bacteroides* sp

Actinomyces *Prevotella* sp.

Propionibacterium *Fusobacterium*

Clostridium sp. (not *C. difficile*)

Other Bacteria – *Pneumocystis carinii*, *Toxoplasmosis gondii*, Malaria

Clindamycin often use for its Antitoxin effect in Toxic Shock Syndrome or Necrotizing Fasciitis due to Group A Strep

Clindamycin

Pharmacology

Absorption – available IV and PO

- Rapidly and completely absorbed ($F = 90\%$); food with minimal effect on absorption

Distribution

- Good serum concentrations with PO or IV
- Good tissue penetration including bone; **minimal CSF penetration**

Elimination

- Clindamycin primarily metabolized by the liver; half-life is 2.5 to 3 hours
- Clindamycin is NOT removed during hemodialysis

Clinical Uses

Anaerobic Infections **OUTSIDE of the CNS**

- Pulmonary, intraabdominal, pelvic, diabetic foot and decubitus ulcer infections

Uncomplicated Skin & Soft Tissue Infections

- Especially in penicillin-allergy

Other

- Alternative for *C. perfringens*, PCP, Toxoplasmosis, malaria, bacterial vaginosis

Clindamycin: Adverse Effects

- Gastrointestinal – 3 to 4 %
 - Nausea, vomiting, diarrhea, dyspepsia
- *C. difficile* colitis – one of worst offenders
 - Mild to severe diarrhea
 - Requires treatment with metronidazole
- Hepatotoxicity - rare
 - Elevated transaminases
- Allergy - rare

Streptogramins

Two Groups

- A: nonpeptide
- B: cyclic peptides

Examples

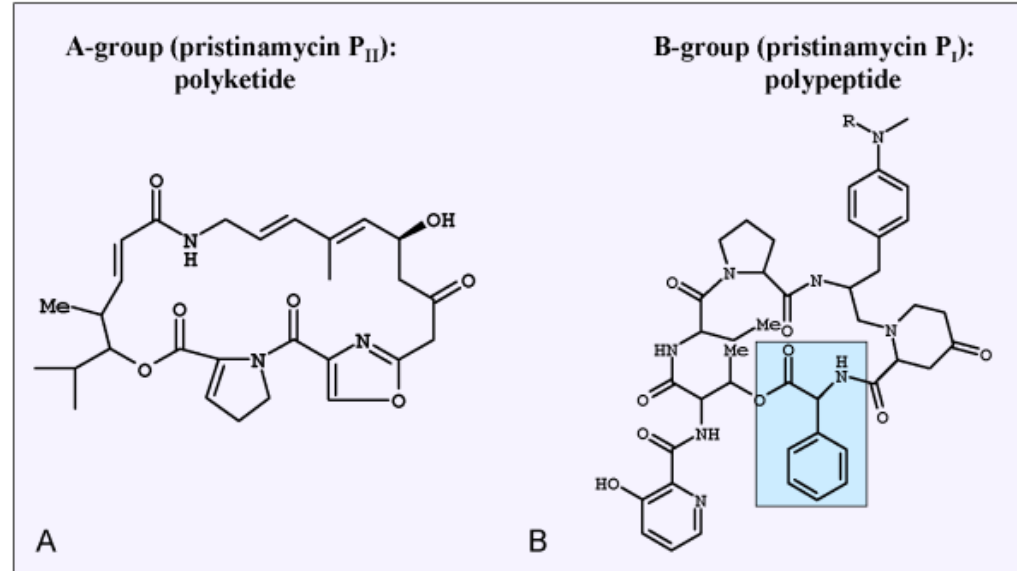
- A: Dalfopristin
- B: Quinupristin

MOA

- Group A
 - Distort ribosomes
 - Prevent tRNA binding
- Group B
 - Block translocation

Results

- Inhibit protein synthesis
- Synergistic if both groups used



- Synercid[®] is the first available agent, which received FDA approval in September 1999
- Developed in response to need for agents with activity **against resistant gram-positives (VRE)**
- **Synercid[®] is a combination of two semi-synthetic pristinamycin derivatives in a 30:70 w/w ratio:**
Quinupristin : Dalfopristin

Synercid®

Mechanism of Action

- Each agent acts on 50S ribosomal subunits to inhibit early and late stages of protein synthesis
- **Bacteriostatic** (cidal against some bacteria)

Mechanism of Resistance

- Alterations in ribosomal binding sites
- Enzymatic inactivation

Synercid® : Spectrum of Activity

Gram-Positive Bacteria

- Methicillin-Susceptible and Methicillin-Resistant *Staph aureus* and coagulase-negative staphylococci
- *Streptococcus pneumoniae* (including PRSP), viridans streptococcus, Group streptococcus
- *Enterococcus faecium* (ONLY)
- *Corynebacterium*, *Bacillus*, *Listeria*, *Actinomyces*
- *Clostridium* sp. (except *C. difficile*), *Peptococcus*, *Peptostreptococcus*

Gram-Negative Aerobes

- Limited activity against *Neisseria* sp. and *Moraxella*

Atypical Bacteria

- *Mycoplasma*, *Legionella*

Synercid®: Clinical Uses

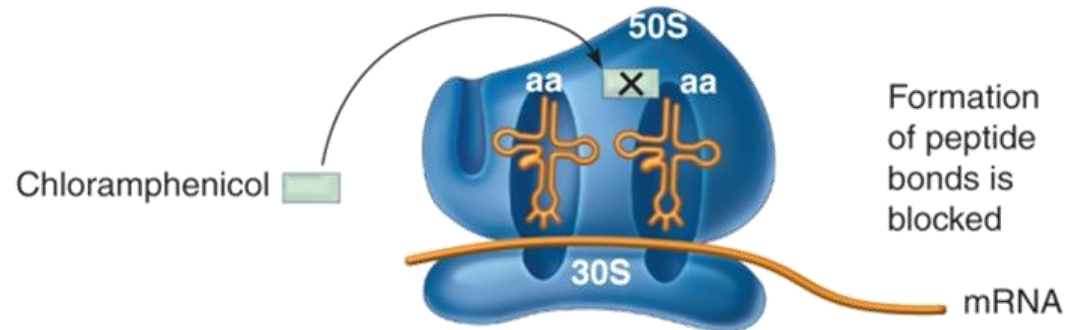
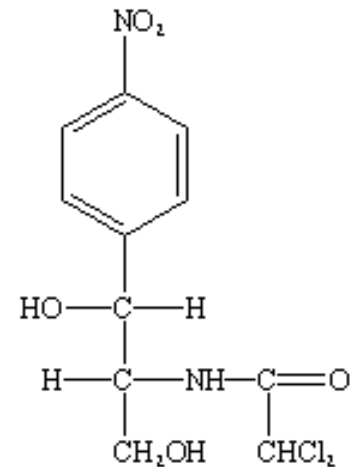
- VRE (*faecium*) bacteremia
- Complicated skin and soft tissue infections due to MSSA or *Streptococcus pyogenes*
- Limited data in treatment of catheter-related bacteremia, infections due to MRSA, and community-acquired pneumonia

Adverse Effects

- **Venous irritation** – especially when administered in peripheral vein (**generally need central line**)
- Gastrointestinal – nausea, vomiting, diarrhea
- **Myalgias, arthralgias** – 2%, Rash
- **↑ total and unconjugated bilirubin**

Chloramphenicol

- Relatively small molecule, easily enters Gram-positive and Gram-negative bacteria
- MOA
 - Attaches to 50S ribosome
 - Interferes with peptide bonds
- Results
 - Inhibits protein synthesis
- Broad spectrum
 - G(+), G(-)
 - Atypical bacteria
 - Anaerobes
 - **Rickettsia**
- PK:
 - **Penetrate well into almost all tissues (CNS)**
 - Elimination
 - Metabolism: Glucoronidation and aryl amine reduction
 - Excretion: Urine (major) and bile (minor)



Chloramphenicol

- Adverse events
 - Bone marrow toxicity
 - Bone marrow suppression (Type A ADR)
 - Aplastic anemia (Type B ADR)
 - Hemolytic anemia in G6PD deficiency
 - Gray baby syndrome
 - N/V and diarrhea
- Drug interaction
 - Potent CYP3A4 and 2C19 inhibitor
- Commonly use empiric treatment of meningitis & veterinary medicine
- Eye drop, eye ointment
 - indications – Eye infection (bacterial conjunctivitis)
- Ear drop
 - indications – ear infection (otitis media)

Aminoglycosides

- Bactericidal inhibitors of protein synthesis
 - Mutations affecting proteins in the bacterial ribosome can confer marked resistance to their action
- Used most widely against gram-negative enteric bacteria
- Drugs:
 - gentamicin, tobramycin, amikacin, netilmicin, kanamycin, streptomycin, paromomycin, and neomycin
 - Streptomycin and Amikacin: mycobacterial infections
 - Paromomycin: intestinal amebiasis

Mechanism of Actions

1. Penetrate bacteria cell wall: passive diffusion
 - Synergist with Penicillins, Vancomycin (↑transport)
2. Transport across cytoplasmic membrane by active transport (proton pump) (oxygen-dependent process)
 - energy-dependent phase I; EDP-I: **rate-limiting step**
 - energy-dependent phase II; EDP-II
3. inhibit protein synthesis :
 - binding irreversibly 16S rRNA of the **30S ribosomal** subunit
 - interfering with bacterial protein translation

Antibacterial activity

- Primarily active against **aerobic gram negative** bacteria
- Gentamicin, tobramycin, and amikacin: directed primarily against aerobic gram-negative bacilli
- Aerobic gram-negative bacilli vary in their susceptibility to the aminoglycosides
 - Anaerobic bacteria are not susceptible!
 - In combination with a cell wall-active agent, such as a penicillin or vancomycin, an aminoglycoside can produce a synergistic bactericidal effect

Aminoglycosides: Spectrum of Activity

- *Gram-Positive Aerobes* (used in combo)

Most *S. aureus* and coagulase-negative staph viridans streptococci

Enterococcus sp. (gentamicin or streptomycin)

- *Gram-Negative Aerobes (not streptomycin)*

E. coli, K. pneumoniae, Proteus sp., Acinetobacter, Citrobacter, Enterobacter sp. Morganella, Providencia, Serratia, Salmonella, Shigella, Pseudomonas aeruginosa (amik>tobra>gent)

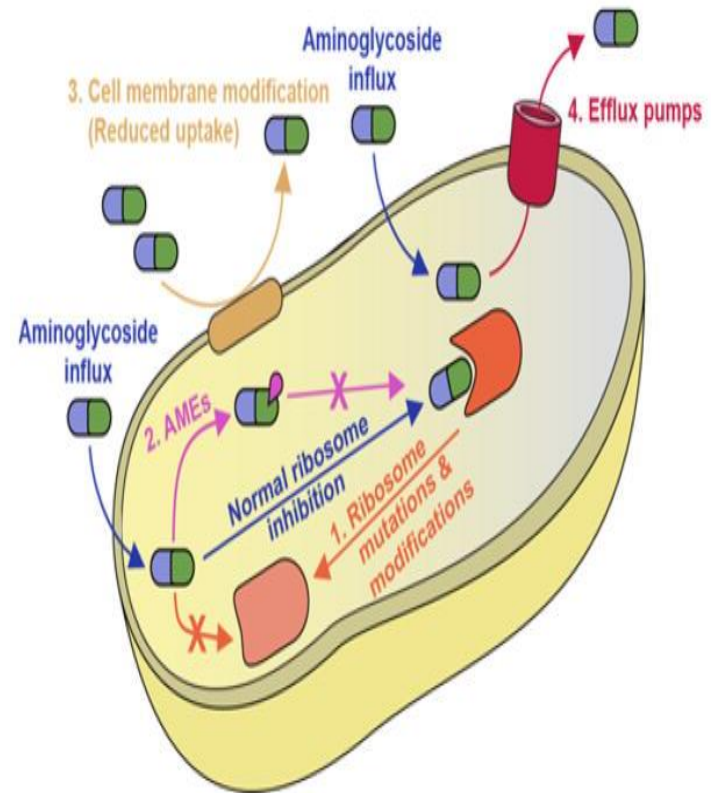
Mycobacteria

—Tuberculosis - streptomycin

—Atypical - streptomycin or amikacin

Mechanisms of resistance

- **Inactivation of the drug by microbial enzymes):** high-level resistance
 - » 1. Aminoglycoside Acetyltransferases
 - » 2. Aminoglycoside phosphotransferases (kinases)
 - » 3. Aminoglycoside nucleotidyltransferases
- **Failure of the antibiotic to penetrate intracellularly**
- **Low affinity of the drug for the bacterial ribosome (modification of the target)**
- **delete/alter of receptor protein on 30s ribosomal unit by mutation at ribosomal proteins or 16S rRNA**



Efflux pump: MexXY
(AmrAB)

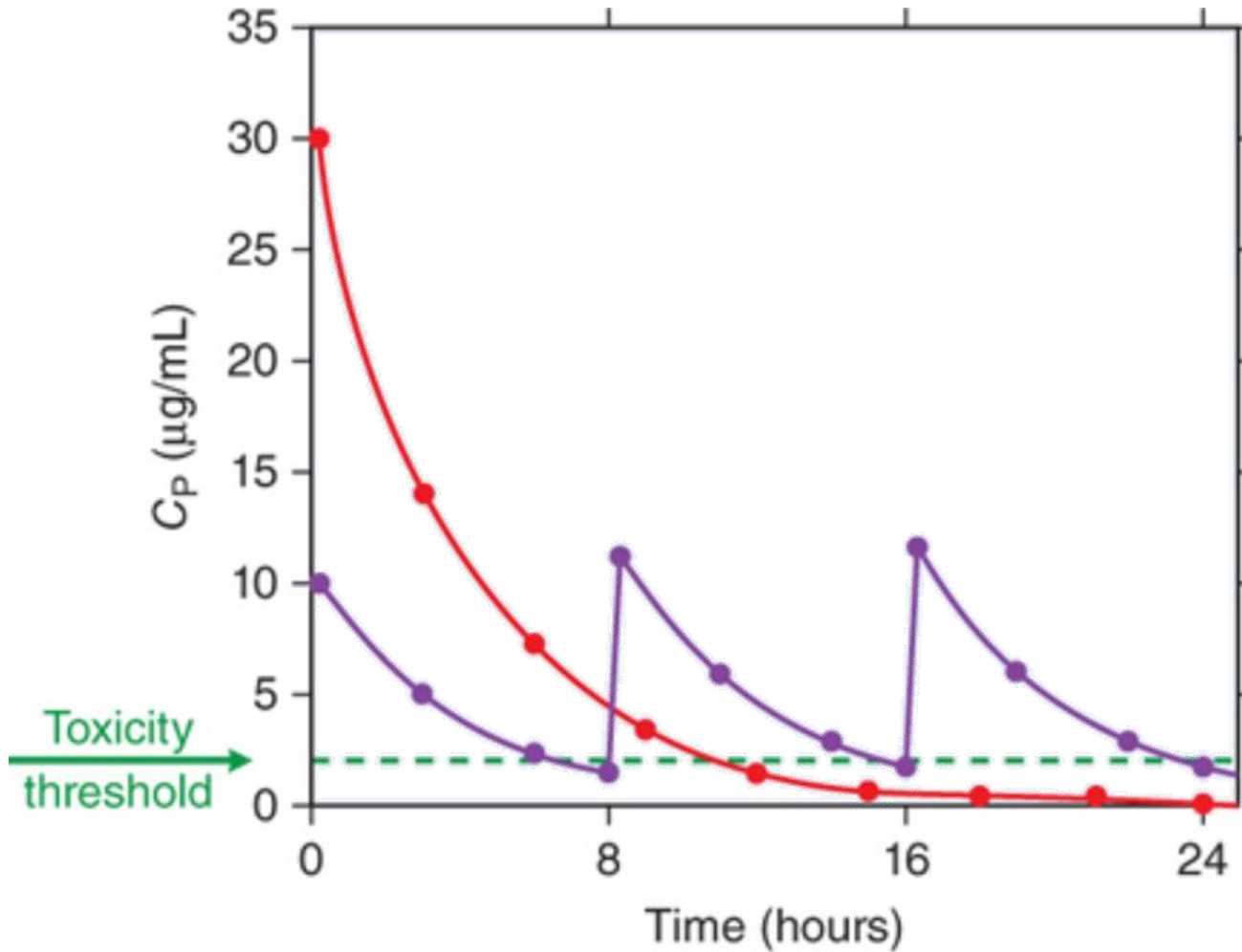
Pharmacokinetics

- **Absorption:** Polar compound, not absorb by oral, administer only by parenteral or local apply (cream, ointment)
- All are absorbed rapidly by *i.m.* injection
- New formulation Inhalation: *P. aeruginosa* pulmonary infections (Tobramycin)
- **Distribution:** Low V_d , poor tissue penetration or blood brain barrier, do not penetrate well into CSF
- **Metabolism and Excretion**
 - Excrete unchanged by renal: direct proportion with creatinine clearance
 - $T_{1/2}$: 2-3 hr. (>100 hr. in patient with renal disease)
 - Patient with renal disease must adjust dose!

Dosing and monitoring

- **1. HIGH-DOSE, EXTENDED-INTERVAL DOSING (“ONCE-DAILY”)**
 - Exhibit equally efficacious and potentially **less toxic** than administration of divided doses
 - Exploits the **concentration-dependent** killing and **post-antibiotic effect** of aminoglycosides
 - High-dose/extended-interval dosing strategy is more controversial include pregnancy, neonates, and pediatrics
 - Safety and efficacy of extended-interval dosing are limited
 - Avoided in patients with significant renal dysfunction (i.e., creatinine clearance < 25 mL/min).

Comparison of single-dose and divided-dose regimens for gentamicin

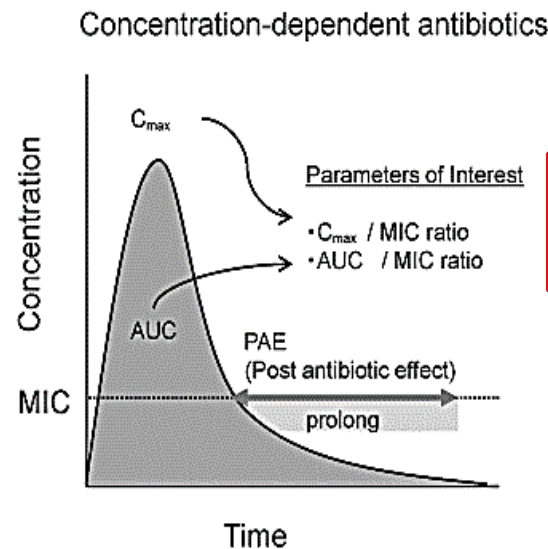
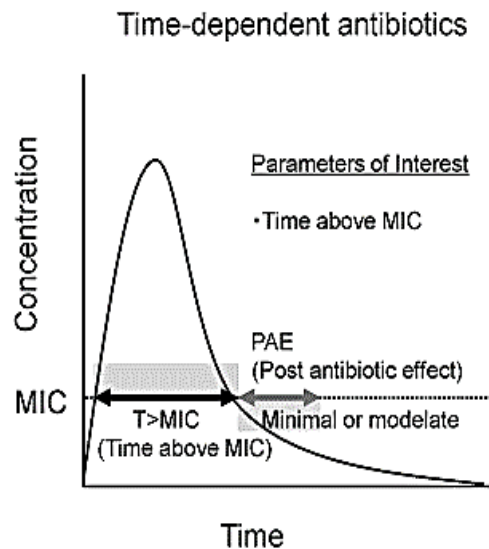


Plasma concentrations are below the threshold for toxicity

- The once-daily regimen: >13-h
- every-8-h regimen: three short (~1 h) periods

Post antibiotic effect (PAE)

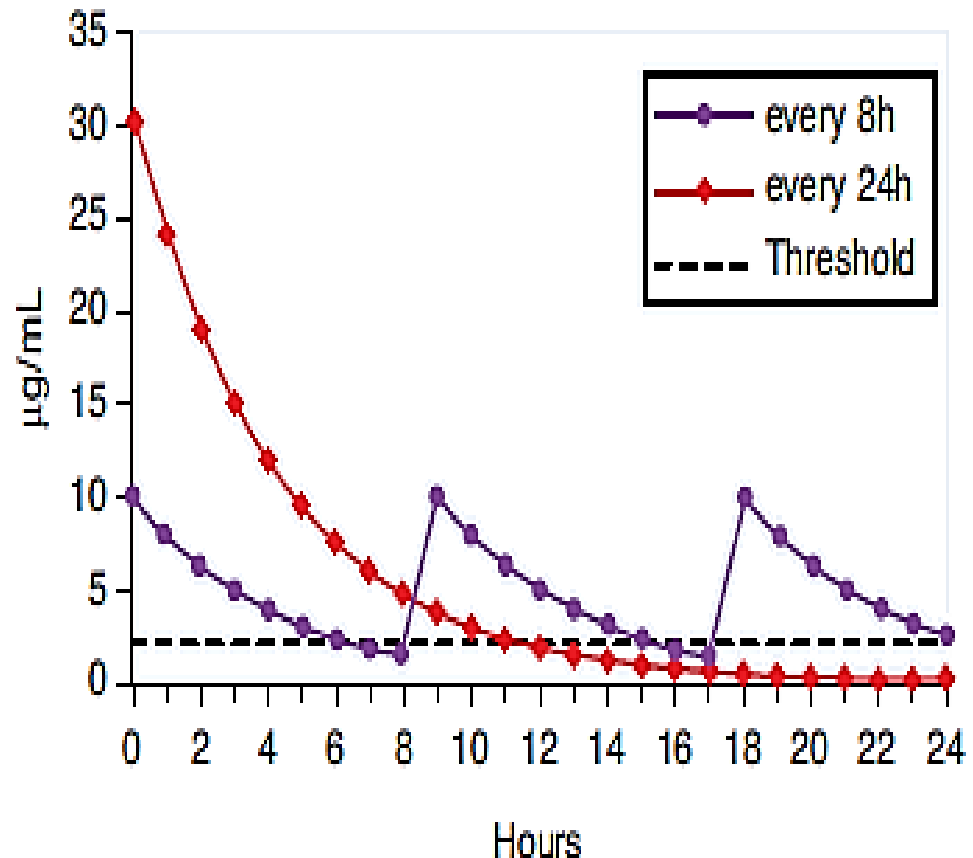
- “Concentration-dependent killing effect”
 - Increase concentration → ↑ killing ability proportion of bacteria and more rapid rate
- “Single daily dose” is more effective, less cost, and less nephrotoxic than multiple doses.



8-10:1 is the best ratio

Once-daily" or "Single-daily" or "pulse" dosing of Aminoglycosides

- 1. Aminoglycosides exhibit a significant post-antibiotic effect (PAE)
- 2. Bactericidal action of aminoglycosides is concentration dependent
- 3. Aminoglycoside uptake into renal tubule cells and the inner ear appears to be saturated at relatively low serum levels
- 4. ↓adaptive post-exposure resistance



High peak conc. : ↑efficacy
Low trough conc.: ↓incidence of nephrotoxicity

Dosing and monitoring (cont.)

- **2. MULTIPLE-DAILY DOSING (“TRADITIONAL”)**
 - **Used when “once-daily” dosing is not appropriate.**
 - Dosing: loading dose and followed by maintenance dose (depend on renal function, CrCl)
 - Both peak and trough plasma concentrations are determined.
 - The peak concentration: therapeutic concentrations
 - The trough concentration: avoid toxicity
 - Monitoring of aminoglycoside plasma concentrations also is important
 - Most accurate method for monitoring plasma levels for dose adjustment: measure the concentration in two plasma samples drawn several hours apart (2 and 12 h after a dose)

Therapeutic uses

- Gentamicin, tobramycin, amikacin, netilmicin: can be used interchangeably for the treatment of most of the infections.
 - Gentamicin is preferred (with long experience and lower cost)
- **Urinary Tract Infections:** gentamicin or tobramycin
- **Pneumonia:** aminoglycoside in combination with a β -lactam antibiotic
- **Meningitis:** use for infections caused by gram-negative organisms resistant to β -lactam antibiotics
- **Peritonitis:** gentamicin, netilmicin, tobramycin, amikacin
- **Bacterial Endocarditis:** “Synergistic” or low-dose gentamicin (3 mg/kg/d) in combination with a penicillin or vancomycin
- **Mycobacterial Infections:** Streptomycin, Amikacin
- **Sepsis:** use only multiple-drug-resistant organism
- **Tularemia:** Streptomycin or gentamicin (1–2 g) is the drug of choice
- **Cystic Fibrosis:** use for multidrug-resistant gram-negative bacilli
- **Topical Applications:** neomycin and paromomycin

Reasons to use in combination with a cell wall-active agents

1. to expand the empiric spectrum of activity of the antimicrobial regimen
2. to provide synergistic bacterial killing
3. to prevent the emergence of resistance to the individual agents

Aminoglycosides: Adverse Effects

1. Nephrotoxicity

- Nonoliguric azotemia due to proximal tubule damage; **increase in BUN and serum Cr**; reversible if caught early
- Risk factors: prolonged high troughs, long duration of therapy (> 2 weeks), underlying renal dysfunction, elderly, other nephrotoxins

2. Ototoxicity

- 8th cranial nerve damage - vestibular and auditory toxicity; irreversible
- Vestibular: dizziness, vertigo, ataxia – Streptomycin, Gentamicin, Tobramycin
- Auditory: tinnitus, decreased hearing – Aminoglycoside, Netilmicin, Gentamicin
- Risk factors: same as for nephrotoxicity

Adverse Effects (cont.)

3. Neuromuscular junction blockade

- Rare incidence but serious!
- Decreased ACh release from NMJ
 - Blockade voltage-gated Ca^{2+} channel
- Curare-like effect: respiratory paralyze
 - Treatment: Calcium (*iv*) or neostigmine(*iv*)
- Neomycin and streptomycin: most often!

4. Hypersensitivity reactions

- Rare incidence
- Skin rash, fever, eosinophilia and anaphylactic shock

Drugs interactions

- 1. Concurrent use of aminoglycoside with other nephrotoxic or ototoxic agents:
↑toxicity
- Ex: Diuretic (loop diuretic), ACEIs, NSAIDs, vancomycin, cyclosporine, cisplatin
- 2. Concurrent use of aminoglycoside with neuromuscular blocking agents: ↑respiratory paralysis
- 3. Simultaneously administered aminoglycoside and penicillin in same line or syringe:
Chemical antagonist (form crystal)

Precaution/Contraindication

- Pregnancy: fetal ototoxicity (category D)
- With other ototoxic drugs: furosemide, minocycline
- With nephrotoxic drugs: vancomycin, cisplatin
- Elderly patients
- Patients with kidney diseases
- Use with other skeletal muscle relaxant

Topical Preparations

- Eye drops & ointment:

- Neomycin, gentamicin, tobramycin

Or

- Combined with steroid:
Dexamethasone
- Ear drops
- Dexamethasone + Neomycin
- Prednisolone + Gentamicin



- Topical formulations:
- Betamethasone + neomycin or gentamicin

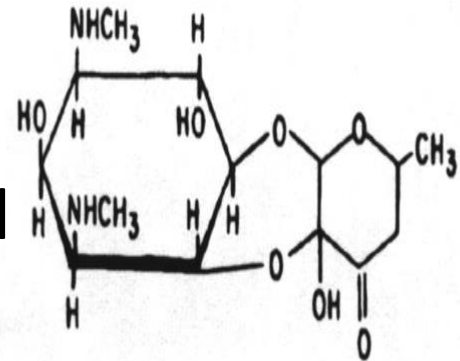


Drug	Therapeutic Uses	Clinical Pharmacology and Tips
Gentamicin (IV)	•UTI •Peritonitis •Endocarditis in combination with a cell-wall active agent •Plague •Tularemia	•Good activity vs. Enterobacteriaceae, Pseudomonas •Some activity vs. Neisseria, Haemophilus, Moraxella •Synergistic activity when combined with a cell-wall agent against many organisms •Vestibular > cochlear toxicity •Toxicity primarily renal and reversible
Tobramycin (IV, inhalation)	•UTI •Lung infections, including cystic fibrosis exacerbations •Nosocomial sepsis of unknown origin	•Similar to gentamicin, with better activity against Pseudomonas aeruginosa •Cochlear ≈ vestibular toxicity
Amikacin (IV)	• UTI • Lung infections, including cystic fibrosis exacerbations • Nosocomial sepsis of unknown origin • Mycobacterial infections	•Similar to tobramycin, with activity against some gram-negative bacilli resistant to other aminoglycosides •Activity against a variety of mycobacteria •Cochlear > vestibular toxicity

Drug	Therapeutic Uses	Clinical Pharmacology and Tips
Streptomycin (IV)	Endocarditis in combination with a cell-wall active agent Tuberculosis Plague Tularemia	Similar to gentamicin , with activity against some gentamicin-resistant enterococci Activity against Mycobacterium tuberculosis Vestibular > cochlear toxicity Vestibular toxicity is irreversible
Neomycin (PO, topical; urologic irrigation)	• Minor skin infections • Bowel preparation prior to intra-abdominal surgery • Bladder irrigation	• Similar activity to gentamicin but only used topically, not systemically • Can cause skin rash
Paromomycin (PO, IM, topical)	• Cryptosporidial infection • Intestinal amebiasis • Leishmaniasis	• Diarrhea, nausea, vomiting • IM use for visceral leishmaniasis • Topical use for cutaneous leishmaniasis

Spectinomycin

- **Aminoglycoside like structure**
- Is aminocyclitol lacking of glycosidic bond and aminosugar
- Bind to 16s rRNA of the 30S ribosomal subunit
 - albeit at a location different from aminoglycoside binding sites
- Parenteral use
- Use only for treatment for gonorrhea in patients who are allergic to penicillin or resistant to other drugs



Spectinomycin. A bacteriostatic aminocyclitol antibiotic that differs from the aminoglycoside group in that it lacks an amino sugar residue.

Summary of Aminoglycoside

- Narrow-spectrum agents
 - Activity limited mainly to gram-negative aerobes
- Prevent protein synthesis by binding 30S ribosome in bacteria
- Recommended for use in combination with a β -lactam to enhance bactericidal effect
- Should be used sparingly and reserved for specific indications
- Duration of therapy and serum concentrations should be monitored to avoid toxicity. (ototoxicity and nephrotoxicity)

Tetracyclines

- Broad Spectrum

- Gram (-), Gram (+)
- Rickettsial organisms

- Generation

- 1st - Tetracycline, Oxytetracycline, Chlortetracycline
- 2nd - Doxycycline (semi-synthetic), Minocycline
- 3rd - Tigecycline

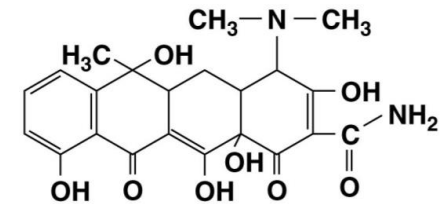
- MOA

- Bind to 30S subunit reversibly
- Block tRNA attachment at A site
- Inhibit codon-anticodon interaction

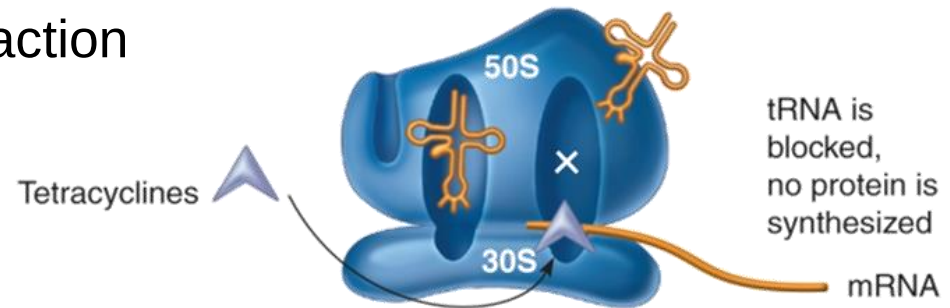
- Widespread resistance

- Side effects

- Affect bone development
- Stain teeth in children

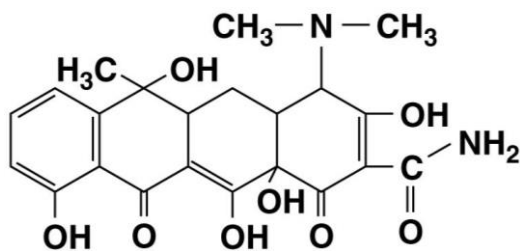


Tetracycline

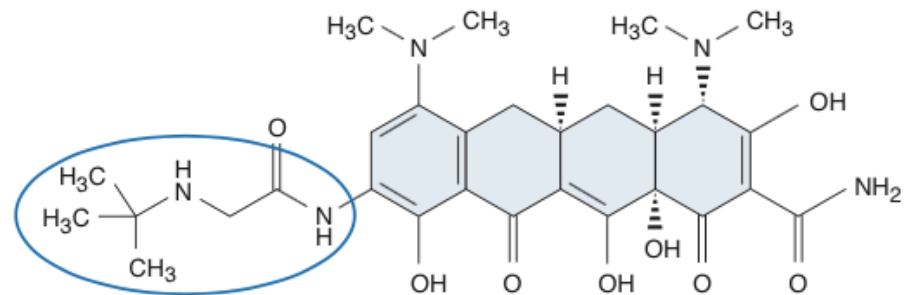


Tetracyclines and Glycylcyclines

- “**Tetracycline**” (discovered 1948) = four linearly six-membered rings
- Agents in use today include **doxycycline**, **minocycline**, tetracycline, demeclocycline and
- Oxytetracycline use as topical, eye ointment, capsule, IV infusion for eye and skin infection
- **Glycylcyclines** are structural modifications **that prevents recognition of tigecycline by many bacterial efflux pumps** and makes it insensitive to modifications of the 30S ribosomal subunit to improve spectrum of activity—tigecycline (Tygacil®) is the only agent available



Tetracycline



Glycyldamido-group

Tigecycline

Tetracyclines

- Mechanism of Action
 - Inhibit bacterial protein synthesis by reversibly binding to the 30S ribosomal subunit
 - Inhibit the binding of aminoacyl transfer-RNA to the acceptor (A) site on the mRNA-ribosomal complex
 - Tetracyclines typically display bacteriostatic activity
- Mechanisms of Resistance
 - Decreased accumulation of tetracycline within bacteria due to decreased permeability or the presence of **efflux**
 - Decreased access of tetracycline to the ribosome due to the presence of **ribosomal protective proteins**
 - Enzymatic inactivation
- Cross resistance is usually observed between the tetracyclines – **except minocycline**

Tetracyclines: Spectrum of Activity

Gram-Positive Aerobes

- *Staphylococcus aureus* (primarily MSSA)
- *Streptococcus pneumoniae* – PSSP (doxycycline – 77 to 88% susceptible)
- Some Group and viridans streptococci
- *Bacillus sp*, *Listeria sp*, *Nocardia sp*

Gram-Negative Aerobes

- *Haemophilus influenzae* (90% susceptible)
- *Haemophilus ducreyi* (chancroid)
- *Campylobacter jejuni*
- *Helicobacter pylori*

Tetracyclines: Spectrum of Activity

Anaerobes

Actinomyces

Propionibacterium

Miscellaneous Bacteria

Bartonella, Bordetella, Brucella, Pasteurella

Legionella, Chlamydia, Mycoplasma, Ureaplasma

Borrelia, Treponema, Leptospira

Rickettsia, Coxiella

Mycobacterium fortuitum

Tigecycline: Spectrum of Activity

Gram-Positive Aerobes

- *Staphylococcus aureus* (MSSA and MRSA)*, Group and viridans streptococci, *Enterococcus faecalis* (VSE), *Listeria sp*

Gram-Negative Aerobes

- *Acinetobacter baumannii*, *Aeromonas hydrophila*, *Citrobacter sp.*, *Escherichia coli*, *Klebsiella sp*, *Serratia marcescens*, *Stenotrophomonas maltophilia*
- NOT *Proteus sp* or *Pseudomonas aeruginosa*

Anaerobes

- *Clostridium perfringens* (difficile), *Bacteroides sp**

Tetracyclines: Pharmacology

Absorption

- Doxycycline is available PO and IV
- Tetracycline and demeclocycline – F = 60 to 80%
- Doxycycline and minocycline - F = 90 to 100%
- Absorption interaction with **di- or trivalent cations**, which may lead to therapeutic failure

Distribution

- Widely distributed with good tissue penetration into synovial fluid, prostate, seminal fluid
- **Minimal CSF penetration**

Tetracyclines: Pharmacology

Elimination

- Demeclocycline and tetracycline are primarily excreted **unchanged in the urine** - require dosage adjustment in the presence of renal insufficiency
- Doxycycline and minocycline are excreted mainly by **non-renal routes** - do not require dosage adjustment in the presence of renal insufficiency
- Tetracyclines are minimally removed during hemodialysis

Clinical Uses of the Tetracyclines

- Community-acquired pneumonia (doxycycline)
- Rickettsial infections – Rocky Mountain Spotted Fever, Q Fever, etc.
- Chlamydial infections
- Acne
- Brucellosis, bartonellosis, plague, tularemia, chancroid, pertussis, anthrax, *H. pylori* (if patient cannot take Clarithromycin, amoxicillin or metronidazole), Lyme disease, etc.
- Syndrome of inappropriate antidiuretic hormone release SIADH (demeclocycline)
- Minocycline is preferable for the treatment of methicillin-resistant staphylococcal infections and also occasionally used for leprosy caused by *Mycobacterium leprae*.

Tetracycline: Adverse Effects

- Gastrointestinal
 - Nausea, vomiting, diarrhea, pseudomembranous colitis
- Hypersensitivity
 - Rash, pruritus, urticaria, angioedema, anaphylaxis
- Photosensitivity
 - Exaggerated sunburn; primarily with demeclocycline
- Renal
 - Fanconi-like syndrome with outdated tetracycline
 - Reversible dose-related diabetes insipidus (demeclocycline)
- Hepatotoxicity
 - Elevated transaminases
- Other

The overall prevalence of staining has been reported to be 3-4% for tetracycline and 3-6% for minocycline.

Prevention of tetracycline or minocycline induced teeth staining or discoloration is most easily done by avoiding tetracycline use during periods of mineralization (or calcification), which generally occurs in children up to the age of 8 years old.

Prevention of minocycline induced staining can be done by avoiding prolonged use, possibly reducing the dose to less than 100 mg per day, and possibly with the coadministration of vitamin C.

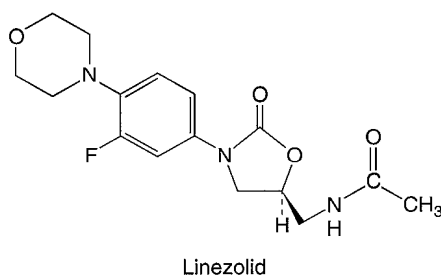
Treatment of tetracycline and minocycline induced staining can be treated with bleaching, composite or porcelain veneers, or crowns, but these methods are only partially effective.

Tigecycline: Clinical Use

- Used for: complicated intraabdominal infections (ESBL), *Acinetobacter baumannii* infection and **carbapenemase-producing strains**, skin/soft tissue infection (MRSA), and sometimes for pneumonia.
- **High rate of failure for HAP/VAP**, use with caution
- Achieves low serum concentrations (distributes widely into tissues) – **poor choice for bacteremia** (also note its static mechanism)
- Main side effects = GI (nausea/vomiting/diarrhea) and elevated LFTs

Oxazolidinones: Linezolid

- Linezolid (Zyvox®) is the first available agent which received FDA approval (2000); available PO and IV
- Developed in response to need for agents with activity against resistant **gram-positives** (MRSA, GISA, VRE)
- Linezolid is a semisynthetic oxazolidinone which is a structural derivative of earlier agents in this class



Oxazolidinones: Linezolid

- Relatively small molecule, easily enters Gram-positive bacteria
- Target is Ribosome
- Linezolid disrupts bacterial growth by inhibiting the initiation process in protein synthesis
- This site of inhibition occurs earlier in the initiation process than other protein synthesis inhibitors (e.g., chloramphenicol, clindamycin, aminoglycosides, and macrolides) that interfere with the elongation process
- Because the site of inhibition is unique to linezolid, cross-resistance to other protein synthesis inhibitors has not yet been reported
- Linezolid may also inhibit virulence factor expression and decrease toxin production in Gram-positive pathogens
- It has been demonstrated that linezolid is bacteriostatic against *Enterococci* and *Staphylococci*, and bactericidal for the majority of *Streptococci*
- Gram-negative bacteria appear to be **naturally resistant**

Oxazolidinones: Linezolid

Mechanism of Action

- Binds to the 50S ribosomal subunit near to surface interface of 30S subunit – causes inhibition of 70S initiation complex which inhibits protein synthesis
- **Bacteriostatic** (cidal against some bacteria); **Since mechanism is static, in general prefer other agents for severe bacteremia, endocarditis.**

Mechanism of Resistance

- Alterations in ribosomal binding sites (RARE) by methylation
- Cross-resistance with other protein synthesis inhibitors is unlikely

Linezolid: Spectrum of Activity

Gram-Positive Bacteria

- Methicillin-Susceptible, Methicillin-Resistant AND Vancomycin-Resistant *Staph aureus* and coagulase-negative staphylococci
- *Streptococcus pneumoniae* (including PRSP), viridans streptococcus, Group streptococcus
- *Enterococcus faecium* AND *faecalis* (including VRE)
- *Bacillus*, *Listeria*, *Clostridium* sp. (except *C. difficile*), *Peptostreptococcus*, *P. acnes*

Gram-Negative Aerobes – relatively inactive

Linezolid

Pharmacology

- Time dependent bactericidal activity
- Absorption – 100% bioavailable
- Distribution – readily distributes into well-perfused tissue; CSF penetration $\approx$ 30%
- Elimination – both renally and nonrenally, but primarily metabolized; $t_{1/2}$ is 4.4 to 5.4 hours; no adjustment for RI; not removed by HD

Clinical Uses

- Use reserved for serious/complicated infections caused by resistant bacteria:
 - VRE bacteremia
 - Complicated skin and soft tissue infections due to MSSA or *Streptococcus pyogenes*
 - CAP due to PSSP or MSSA
 - Nosocomial pneumonia due to MSSA or MRSA
- Limited data in treatment of serious infections due to MRSA or VRE (endocarditis, meningitis, osteomyelitis); catheter-related bacteremia

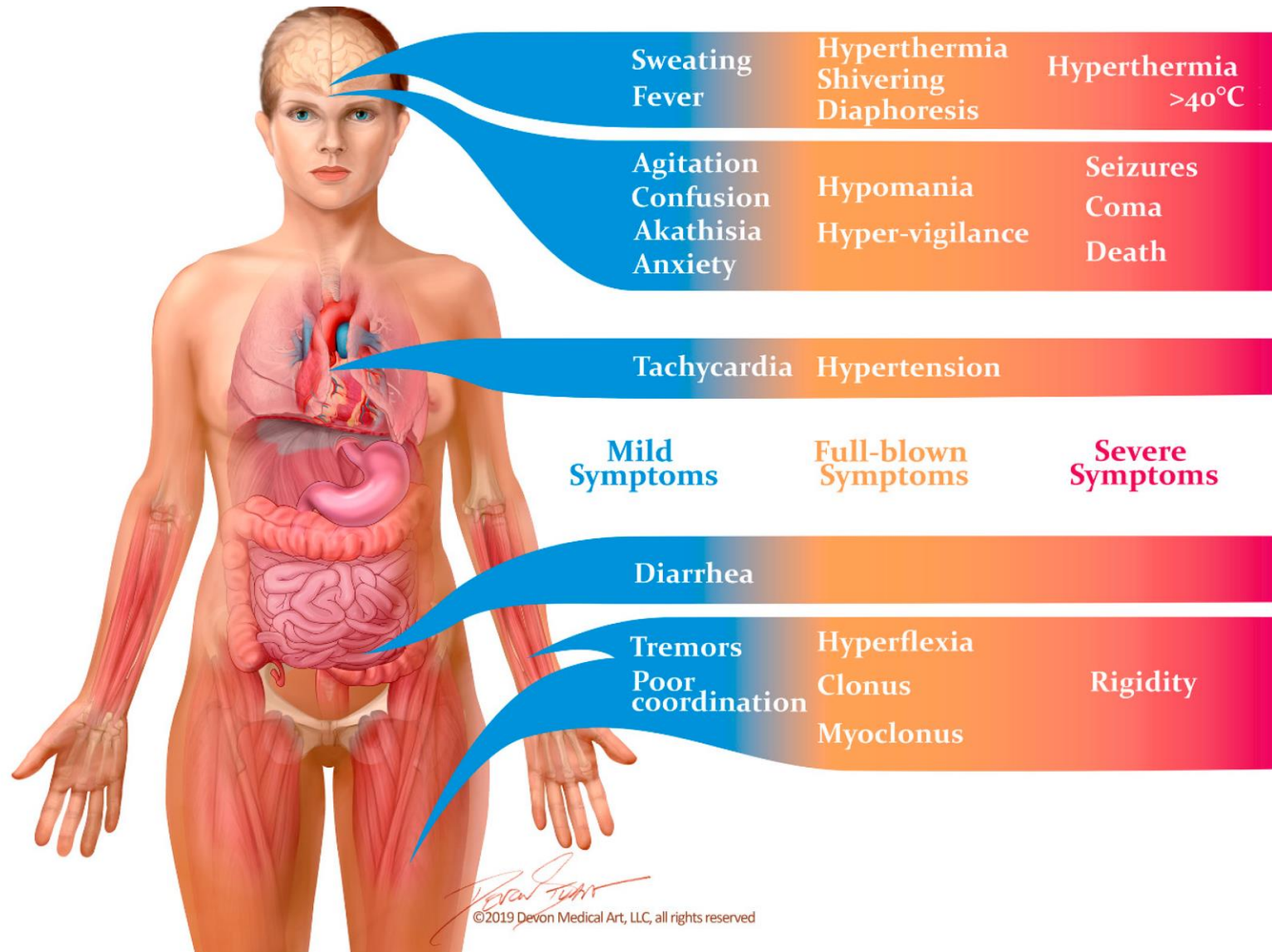
Also: may be superior vs Vancomycin for complicated skin/soft tissue infections due to MRSA – similar rationale (tissue penetration, antitoxin effect)

Linezolid: Adverse Effects

- Gastrointestinal – N/V, diarrhea (6 to 8 %)
- Headache – 6.5%
- Long term usage can lead to ***mitochondrial toxicity*** à lactic acidosis, peripheral neuropathy, optic neuritis and blindness.
- **Serotonin syndrome** - linezolid inhibits the action of monoamine oxidase A—an enzyme responsible for breaking down serotonin in the brain
- **Bone marrow suppression:**
 - Thrombocytopenia – 2 to 4%
 - Most often with treatment durations of > 2 weeks
 - Therapy should be discontinued – platelet counts will return to normal
- **In general, Linezolid is well tolerated for short-term, but not so much long-term.**

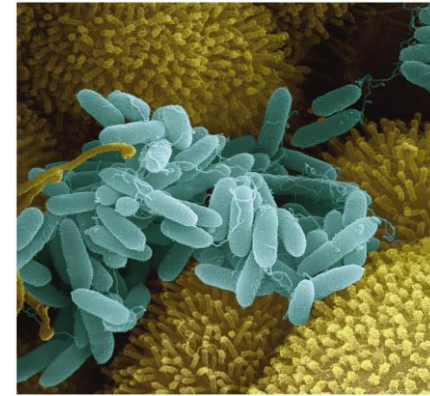
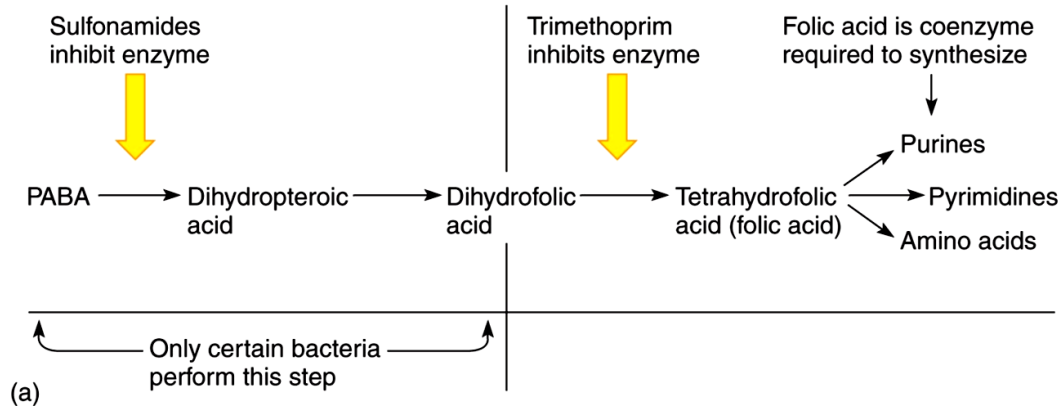
Linezolid: Adverse Effects

Serotonin syndrome



Antibiotic Drugs

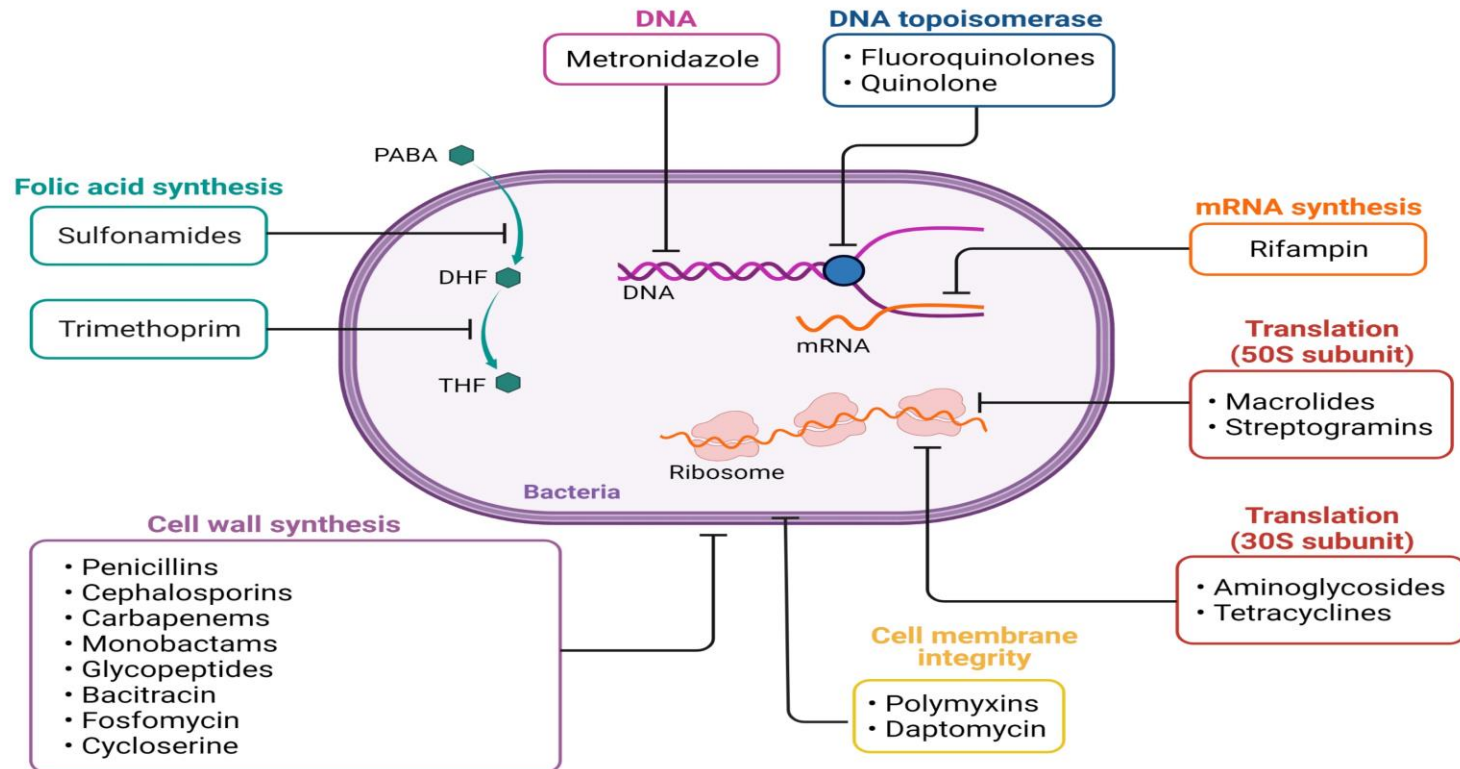
5. Metabolite Inhibition



Folate
Synthesis
Inhibitors

Provide sequential inhibition of folinic acid synthesis which is necessary for microbial production of DNA

- **Sulfamethoxazole** inhibits **dihydropteroate synthase** – inhibits incorporation of p-aminobenzoic acid (PABA) into dihydrofolate
- **Trimethoprim** inhibits **dihydrofolate reductase** – prevents reduction of dihydrofolate to tetrahydrofolate
- Each agent alone is bacteriostatic, however, the combination displays bactericidal activity



TMP-SMX

Mechanisms of Resistance

- Develops more slowly to the combination as opposed to either agent alone
- Reported in *E. coli*, *Klebsiella sp*, *Proteus sp*, and *H. influenzae*
- Mediated by point mutations in dihydro-pterolate synthase and/or altered production or sensitivity of dihydrofolate reductase

TMP-SMX: Spectrum of Activity

Gram-Positives

Some *S. pneumoniae*

Staph aureus

S. pyogenes

Nocardia

Anaerobes

No Activity

Other

Pneumocystis carinii

Gram-Negatives

Acinetobacter

Enterobacter

E. coli

K. pneumoniae

Proteus

Salmonella, Shigella

Haemophilus sp.

N. gonorrhoeae

Stenotrophomonas maltophilia

TMP-SMX: Pharmacology

- Optimal synergistic ratio against bacteria in serum and tissue is 1:20 (TMP:SMX) – **achieved by administering TMP-SMX in fixed oral or IV dose in a 1:5 ratio**
- Absorption – available IV and PO
 - Rapidly and completely absorbed ($F > 90\%$)
 - Peaks are higher and more predictable with IV administration
- Distribution – urine, prostate, CSF, CNS
- Metabolisms – Acetylation (major), Glucuronidation (minor)
- Elimination – dual; requires dosage adjustment when $\text{CrCl} < 30 \text{ ml/min}$

TMP-SMX: Clinical Uses

- Acute, chronic, or recurrent infections of the urinary tract (UTI)
- Acute or chronic bacterial prostatitis
- Acute bacterial exacerbations of chronic bronchitis (ABECB)
- *Pneumocystis carinii* pneumonia – DRUG OF CHOICE for treatment and prophylaxis
- *Salmonella*, *Shigella*, traveler's diarrhea (gastroenteritis)
- *Nocardia*, *Stenotrophomonas*, *Toxoplasmosis*

TMP-SMX: Adverse Effects

- Gastrointestinal
 - Nausea, vomiting, diarrhea, glossitis
- Hematologic
 - Bone marrow suppression: Leukopenia, thrombocytopenia, eosinophilia
 - G6PD deficiency (hemolytic anemia)
 - Aplastic anemia
- Hypersensitivity
 - Rash, urticaria, toxic epidermal necrolysis, Steven's-Johnson
- CNS
 - Headache, aseptic meningitis, seizures
- Renal
 - increased creatinine (both from pseudo-Cr elevation due to blocked Cr secretion into tubules – average increase in Cr by ~20%, and true AKI from Interstitial Nephritis and ATN), hyperkalemia (esp high doses and with CKD)
- Other – crystalluria, hematuria and urinary constriction, kernicterus in newborn